In [1]:
```
from sklearn import preprocessing
```

In [2]:
```
import pandas as pd
```

In [3]:
```
import numpy as np
```

In [5]:
```
df=pd.read_csv("iris.csv")
```

In [10]:
```
col_names = ['SepalLength', 'SepalWidth', 'PetalLength', 'PetalWidth']
```

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Id           150 non-null    int64
 1   SepalLength  148 non-null    float64
 2   SepalWidth   150 non-null    float64
 3   PetalLength  150 non-null    float64
 4   PetalWidth   148 non-null    float64
 5   Species      150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 7.2 KB
```

In [12]:
```
min_max=preprocessing.MinMaxScaler()
```

In [13]:
```
df[col_names]=min_max.fit_transform(df[col_names])
df
```

Out[13]:

	Id	SepalLength	SepalWidth	PetalLength	PetalWidth	Species
0	1	0.222222	0.625000	0.067797	0.041667	Setosa
1	2	0.166667	0.416667	0.067797	0.041667	Setosa
2	3	0.111111	0.500000	0.050847	0.041667	Setosa
3	4	0.083333	0.458333	0.084746	0.041667	Setosa
4	5	0.194444	0.666667	0.067797	0.041667	Setosa
...	...	...	...	...	...	...
145	146	0.666667	0.416667	0.711864	0.916667	Virginica
146	147	0.555556	0.208333	0.677966	0.750000	Virginica
147	148	0.611111	0.416667	0.711864	0.791667	Virginica
148	149	0.527778	0.583333	0.745763	0.916667	Virginica
149	150	0.444444	0.416667	0.694915	0.708333	Virginica

150 rows × 6 columns

In [15]:
```
df['Species'].unique()
```

Out[15]:
```
<StringArray>
['Setosa', 'Versicolor', 'Virginica']
Length: 3, dtype: str
```

In [16]:
```
lable_enc=preprocessing.LabelEncoder()
```

In [19]:
```
df['Species']=lable_enc.fit_transform(df['Species'])
```

In [20]:
```
df
```

Out[20]:

	Id	SepalLength	SepalWidth	PetalLength	PetalWidth	Species	species
0	1	0.222222	0.625000	0.067797	0.041667	0	0
1	2	0.166667	0.416667	0.067797	0.041667	0	0
2	3	0.111111	0.500000	0.050847	0.041667	0	0
3	4	0.083333	0.458333	0.084746	0.041667	0	0
4	5	0.194444	0.666667	0.067797	0.041667	0	0
...	...	...	...	...	...	...	...
145	146	0.666667	0.416667	0.711864	0.916667	2	2
146	147	0.555556	0.208333	0.677966	0.750000	2	2
147	148	0.611111	0.416667	0.711864	0.791667	2	2
148	149	0.527778	0.583333	0.745763	0.916667	2	2
149	150	0.444444	0.416667	0.694915	0.708333	2	2

150 rows × 7 columns

In [21]:
```
one_hot=preprocessing.OneHotEncoder()
```

In [22]:
```
encoded=one_hot.fit_transform(df[['species']]).toarray()
```

In [23]:
```
df
```

Out[23]:

	Id	SepalLength	SepalWidth	PetalLength	PetalWidth	Species	species
0	1	0.222222	0.625000	0.067797	0.041667	0	0
1	2	0.166667	0.416667	0.067797	0.041667	0	0
2	3	0.111111	0.500000	0.050847	0.041667	0	0
3	4	0.083333	0.458333	0.084746	0.041667	0	0
4	5	0.194444	0.666667	0.067797	0.041667	0	0
...	...	...	...	...	...	...	...
145	146	0.666667	0.416667	0.711864	0.916667	2	2
146	147	0.555556	0.208333	0.677966	0.750000	2	2
147	148	0.611111	0.416667	0.711864	0.791667	2	2
148	149	0.527778	0.583333	0.745763	0.916667	2	2
149	150	0.444444	0.416667	0.694915	0.708333	2	2

150 rows × 7 columns

In [24]:
```
encoded_df=pd.DataFrame(encoded)
```

In [25]:
```
df
```

Out[25]:

	Id	SepalLength	SepalWidth	PetalLength	PetalWidth	Species	species
0	1	0.222222	0.625000	0.067797	0.041667	0	0
1	2	0.166667	0.416667	0.067797	0.041667	0	0
2	3	0.111111	0.500000	0.050847	0.041667	0	0
3	4	0.083333	0.458333	0.084746	0.041667	0	0
4	5	0.194444	0.666667	0.067797	0.041667	0	0
...	...	...	...	...	...	...	...
145	146	0.666667	0.416667	0.711864	0.916667	2	2
146	147	0.555556	0.208333	0.677966	0.750000	2	2
147	148	0.611111	0.416667	0.711864	0.791667	2	2
148	149	0.527778	0.583333	0.745763	0.916667	2	2
149	150	0.444444	0.416667	0.694915	0.708333	2	2

150 rows × 7 columns

In [26]:

```
encoded_df
```

Out[26]:

	0	1	2
0	1.0	0.0	0.0
1	1.0	0.0	0.0
2	1.0	0.0	0.0
3	1.0	0.0	0.0
4	1.0	0.0	0.0
...	...	...	...
145	0.0	0.0	1.0
146	0.0	0.0	1.0
147	0.0	0.0	1.0
148	0.0	0.0	1.0
149	0.0	0.0	1.0

150 rows × 3 columns

In [27]:

```
encoded
```

Out[27]:
```
array([[1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [
```

```
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
```

```
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 1., 0.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
[0., 0., 1.],
```

```
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.]])
```

In [ ]: